

# Fluid Coherence and Blow-up Diagnostics

A MAAT Early-Warning Benchmark for Burgers and Navier–Stokes Toy Flows

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## Abstract

This paper tests whether MAAT-style structural supports can be used as early-warning diagnostics for fluid steepening and turbulence-like activity. The goal is deliberately modest: to test whether defect-field supports can distinguish a controlled finite-gradient warning case from a regular fluid-control case.

Two toy systems are used. First, a one-dimensional viscous Burgers equation is evolved from smooth initial data with a known inviscid shock time. Second, an unforced two-dimensional incompressible Navier–Stokes vorticity simulation is used as a regular control. Operational supports measure equation residual consistency, energy or enstrophy balance, controlled activity pressure, spectral coherence, and the v1.2.1 robustness closure.

In the Burgers benchmark, the MAAT warning functional correlates strongly with gradient steepening: Spearman correlation with  $\max |u_x|$  is 0.8413, and the pre-shock correlation with inverse time-to-shock is 0.9952. The warning crosses a high-alert threshold at  $t \simeq 1.16$ , before the inviscid shock time  $t_* \simeq 1.3334$ . This positive Burgers response is partly expected because the diagnostic explicitly includes activity pressure and spectral-tail growth. In the two-dimensional Navier–Stokes control, the warning remains bounded with zero high-warning events under the chosen threshold.

The main contribution is therefore the controlled alarm/no-alarm contrast: MAAT fluid supports may be useful as structural early-warning diagnostics, not as a proof mechanism for fluid regularity.

## 1 Purpose

The Navier–Stokes equations are a natural stress test for any structural selection framework. Their central qualitative question is not merely whether a solution satisfies an equation at one time, but whether coherent flow structure persists under nonlinear transport, vortex stretching, dissipation, and energy transfer.

For MAAT, this suggests a diagnostic question:

*Can structural defect fields warn when a fluid trajectory is drifting toward loss of coherence?*

This question must be separated from the Clay Millennium problem. The present paper is a toy diagnostic benchmark. The test is whether a MAAT-like warning functional responds to controlled gradient steepening while remaining bounded in a regular two-dimensional control flow.

In the language of MAAT v1.6, the supports are not only static descriptors. They serve as time-dependent navigation coordinates along an evolution trajectory:

$$\Phi_0 \rightarrow \Phi_1 \rightarrow \Phi_2 \rightarrow \cdots .$$

The warning functional asks whether the trajectory is entering a region of increasing structural stress.

## 2 Structural Fluid Supports

For a field trajectory  $\Phi(t)$  obeying a fluid equation, define operational defects:

$$d_H = \frac{\|\partial_t \Phi - \mathcal{N}[\Phi]\|_{\text{rms}}}{\|\mathcal{N}[\Phi]\|_{\text{rms}} + \epsilon}, \quad (1)$$

$$d_B = \frac{|\dot{E} + \mathcal{D}|}{|\mathcal{D}| + \epsilon}, \quad (2)$$

where  $\mathcal{N}[\Phi]$  is the equation right-hand side,  $E$  is the relevant energy or enstrophy, and  $\mathcal{D}$  is the corresponding dissipative term.

The supports are:

$$H = \frac{1}{1 + d_H}, \quad B = \frac{1}{1 + d_B}, \quad (3)$$

$$S_{\text{eff}}(A) = \exp \left[ -\frac{1}{2} \left( \frac{\log A - \log A_\star}{\sigma_A} \right)^2 \right], \quad V = \text{low-mode spectral coherence}. \quad (4)$$

Here  $A$  is an activity pressure: for Burgers it is normalised gradient enstrophy, while for two-dimensional Navier–Stokes it is normalised palinstrophy. The activity support is intentionally bounded: large activity is not automatically interpreted as healthy structure.

The robustness closure is:

$$R_{\text{resp}} = (HBV)^{1/3}, \quad R_{\text{rob}} = \min \left( R_{\text{resp}}, (HBS_{\text{eff}}V)^{1/4} \right). \quad (5)$$

The warning functional is:

$$\boxed{W_{\text{MAAT}} = (1 - R_{\text{rob}})\sqrt{A}(1 + \Theta_{\text{tail}})}, \quad (6)$$

where  $\Theta_{\text{tail}}$  measures high-frequency spectral tail pressure. Thus the warning grows when activity concentrates while robustness and spectral coherence degrade.

## 3 Test Systems

### 3.1 Burgers Shock-Warning Proxy

The first test uses the viscous Burgers equation

$$u_t + uu_x = \nu u_{xx} \quad (7)$$

on a periodic domain. The inviscid limit develops a shock when

$$t_* = -\frac{1}{\min_x u_x(x, 0)}. \quad (8)$$

This provides a controlled finite-gradient warning target. The simulation uses a pseudo-spectral semi-implicit scheme with  $N = 256$  modes and  $\nu = 5 \times 10^{-4}$ .

### 3.2 Two-Dimensional Navier–Stokes Control

The second test uses the incompressible two-dimensional vorticity equation

$$\omega_t + u \cdot \nabla \omega = \nu \Delta \omega, \quad u = \nabla^\perp \psi, \quad \omega = -\Delta \psi. \quad (9)$$

This is not a blow-up benchmark. Two-dimensional Navier–Stokes is used here as a regularity/control case: the diagnostic may track activity, but it should not generate false singularity-like alarms in a decaying unforced flow.

## 4 Results

Table 1: Summary of Paper-60 fluid diagnostic benchmark.

| Quantity                                                       | Value  |
|----------------------------------------------------------------|--------|
| Burgers inviscid shock time                                    | 1.3334 |
| Burgers first 70% warning time                                 | 1.1600 |
| Burgers warning lead time                                      | 0.1734 |
| Spearman $W_{\text{MAAT}}$ vs. $\max  u_x $                    | 0.8413 |
| Pre-shock Spearman $W_{\text{MAAT}}$ vs. inverse time-to-shock | 0.9952 |
| 2D Navier–Stokes false high-warning fraction                   | 0.0000 |
| 2D Navier–Stokes minimum $R_{\text{rob}}$                      | 0.8956 |

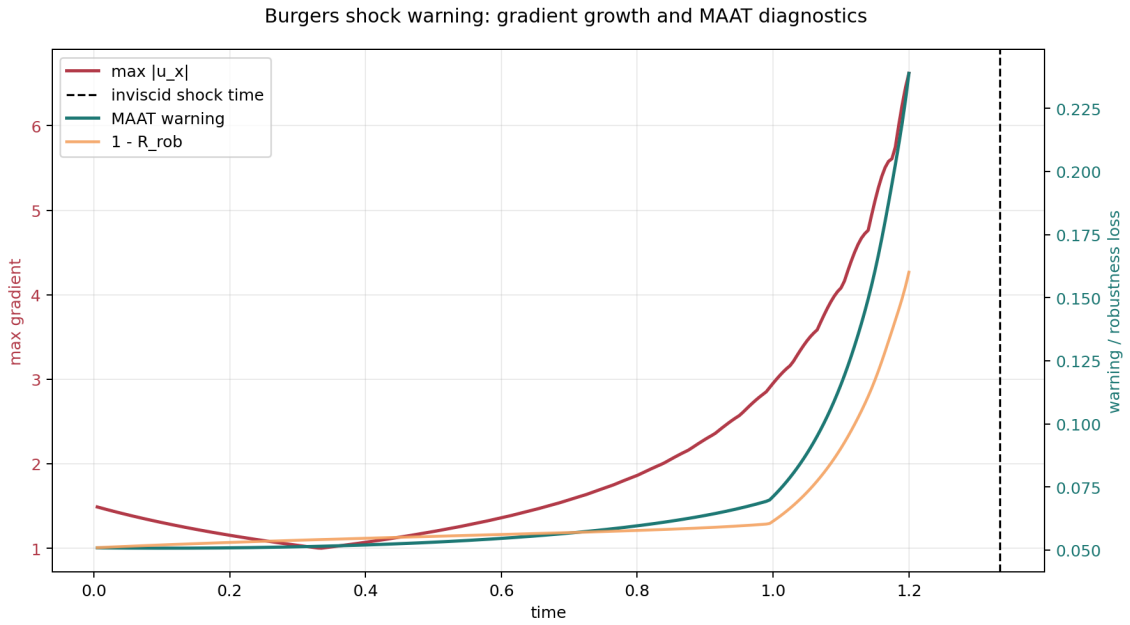


Figure 1: Burgers shock-warning proxy. The MAAT warning functional rises with gradient steepening and crosses a high warning threshold before the inviscid shock time.

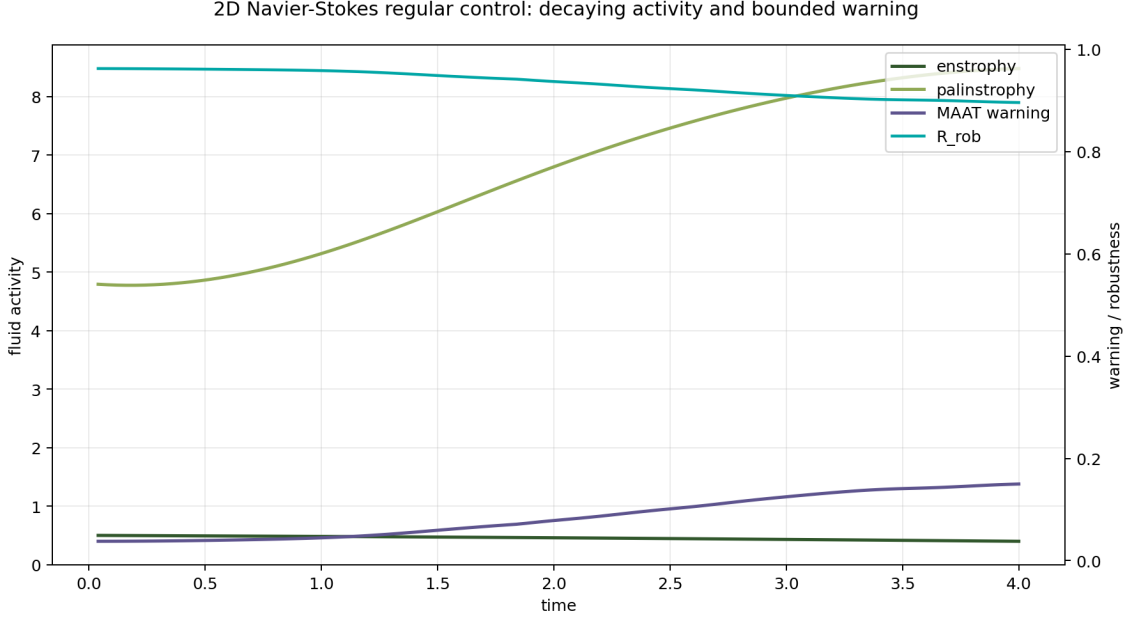


Figure 2: Two-dimensional Navier–Stokes regular-control case. The warning remains bounded while enstrophy decays and palinstrophy increases moderately.

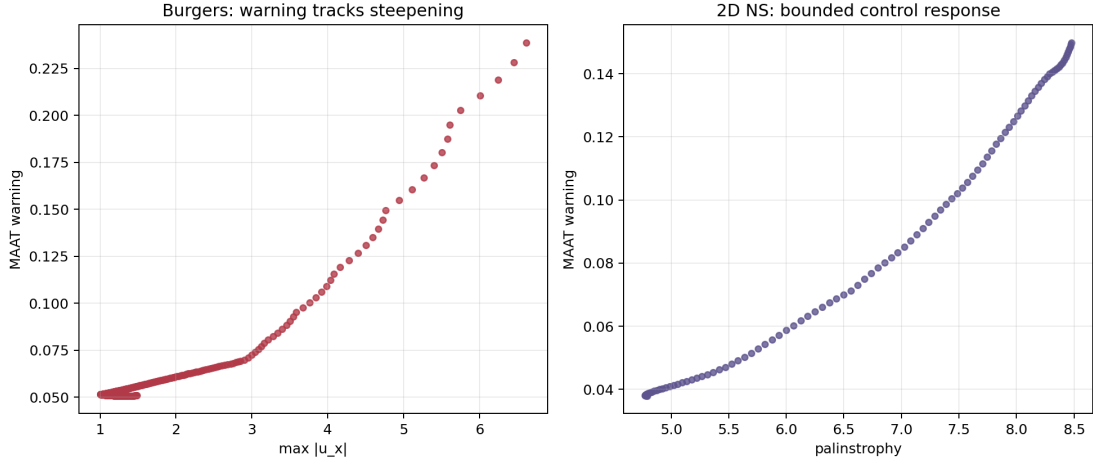


Figure 3: Warning scatter plots. The Burgers warning tracks steepening; the two-dimensional control remains a bounded activity response rather than a blow-up alarm.

## 5 Interpretation

The positive result is not that MAAT solves a PDE problem. The positive result is narrower: the same support language can distinguish two structurally different fluid situations. In Burgers, activity pressure and high-frequency tail growth are built into the warning design, so the correlation with gradient steepening is not surprising by itself. In the two-dimensional Navier–Stokes control, activity is present but remains structurally bounded. This no-false-alarm behaviour is the more important control result: the diagnostic responds to steepening without simply interpreting all fluid activity as pathological.

This is exactly the type of role MAAT can plausibly play in fluid dynamics:

*a structural early-warning layer over standard PDE evolution.*

Such a layer could be useful for numerical monitoring, turbulence diagnostics, or identifying candidate regions of high structural stress in more expensive three-dimensional simulations.

## 6 Limitations

- The Burgers equation is a shock proxy, not Navier–Stokes.
- The Navier–Stokes simulation is two-dimensional and unforced; it is not a three-dimensional blow-up test.
- The supports are heuristic operational diagnostics, not derived from a fluid variational principle.
- The warning threshold is diagnostic and not calibrated against physical data.
- A serious next step would test forced turbulence, vortex-stretching proxies, adaptive mesh simulations, or public direct-numerical-simulation datasets.

## 7 Reproducibility

The experiment is stored in:

```
experiments/fluid_blowup_diagnostics_paper60/
```

Run:

```
cd experiments/fluid_blowup_diagnostics_paper60
python3 fluid_blowup_diagnostics_paper60.py
```

Outputs are written to:

```
outputs_paper60/
```

The public repository is:

<https://github.com/Chris4081/structural-selection-principle>

No external fluid dataset is redistributed. All fields are generated locally by the script from synthetic initial conditions. CSV, JSON, and PNG files are derived reproducibility artifacts.

## 8 Conclusion

Navier–Stokes is structurally natural for MAAT because it asks whether coherent flow survives nonlinear transport and cascade pressure. The present paper takes the first safe step: it builds a reproducible early-warning diagnostic and tests it against one finite-gradient proxy and one regular-control flow.

The result is encouraging but bounded. MAAT fluid supports can track loss of structural coherence in a Burgers shock-warning setting and remain bounded in a two-dimensional Navier–Stokes control. This motivates future work on three-dimensional turbulence diagnostics, but it does not provide a regularity proof.

## References

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